# Capturing the Essence of a Phrase: Extracting Physical and Sensory Information from Text


**Abhinav Gupta, Jesse Thomason**

GLAMOR Lab, University of Southern California, Los Angeles, CA, USA

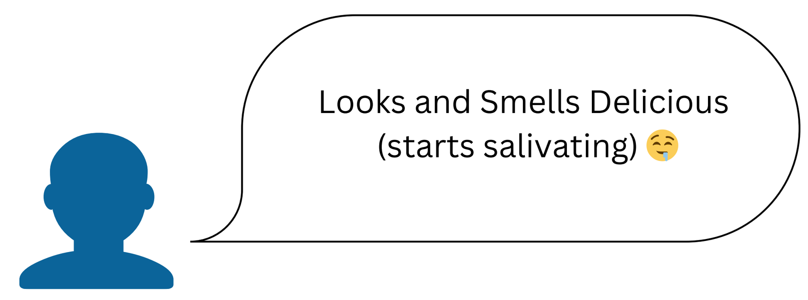

✨GLAMOR

USC Viterbi School of Engineering

## 1. Introduction

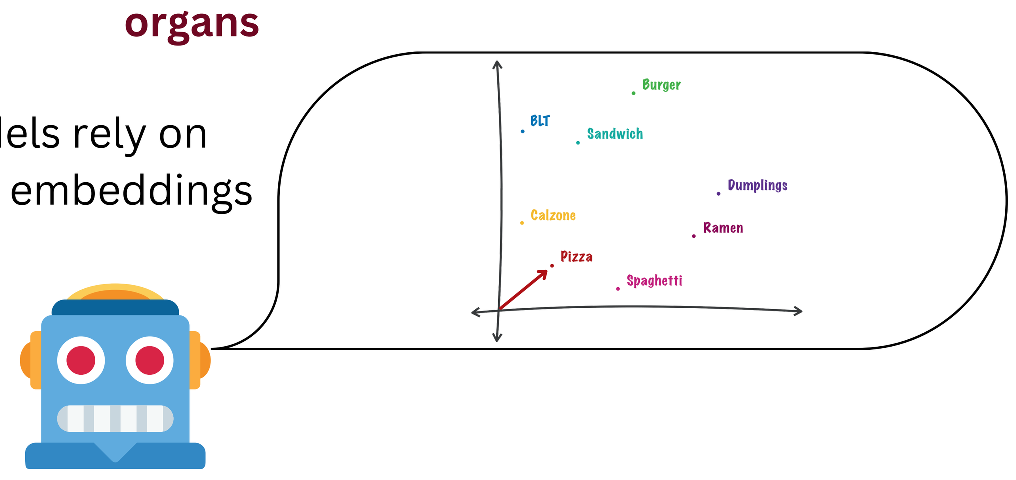
Word Embeddings (**Word2Vec, GloVe, BERT**) can capture **semantic and syntactic relationships** between words but do not give us any **qualitative information** about the words they encode.

What do you think of "Pizza 🍕"?




**Human cognition relies on the sensory and motor inputs from various organs**

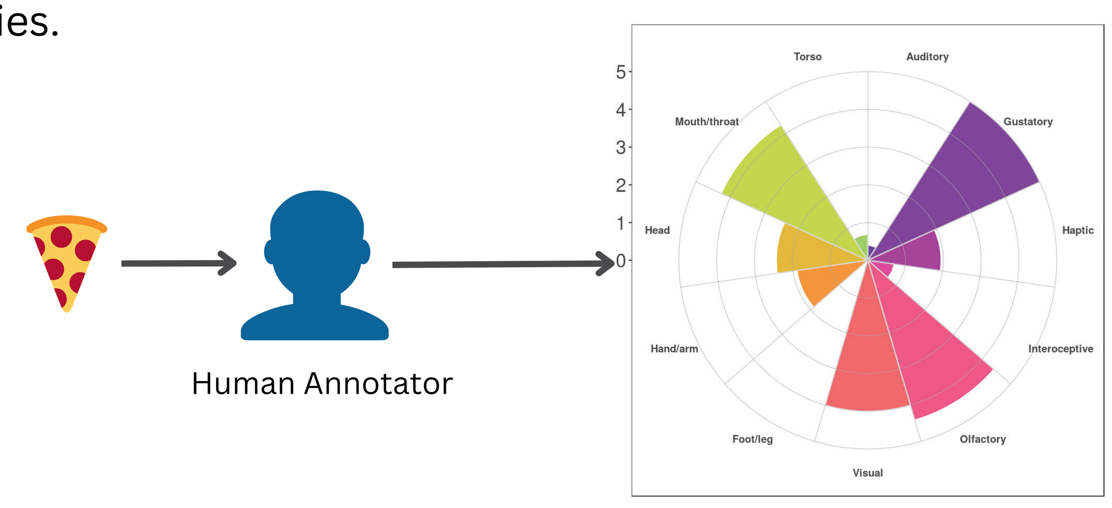
Meanwhile, language models rely on **lexical** and **contextual** word embeddings




**Can we extract sensorimotor information from word embeddings?**

## 2. Lancaster Sensorimotor Norms

Lancaster Sensorimotor Norms are **multidimensional measures** of **perceptual and action strength** for 40,000 English words across 11 sensorimotor modalities.




Sensorimotor Norms

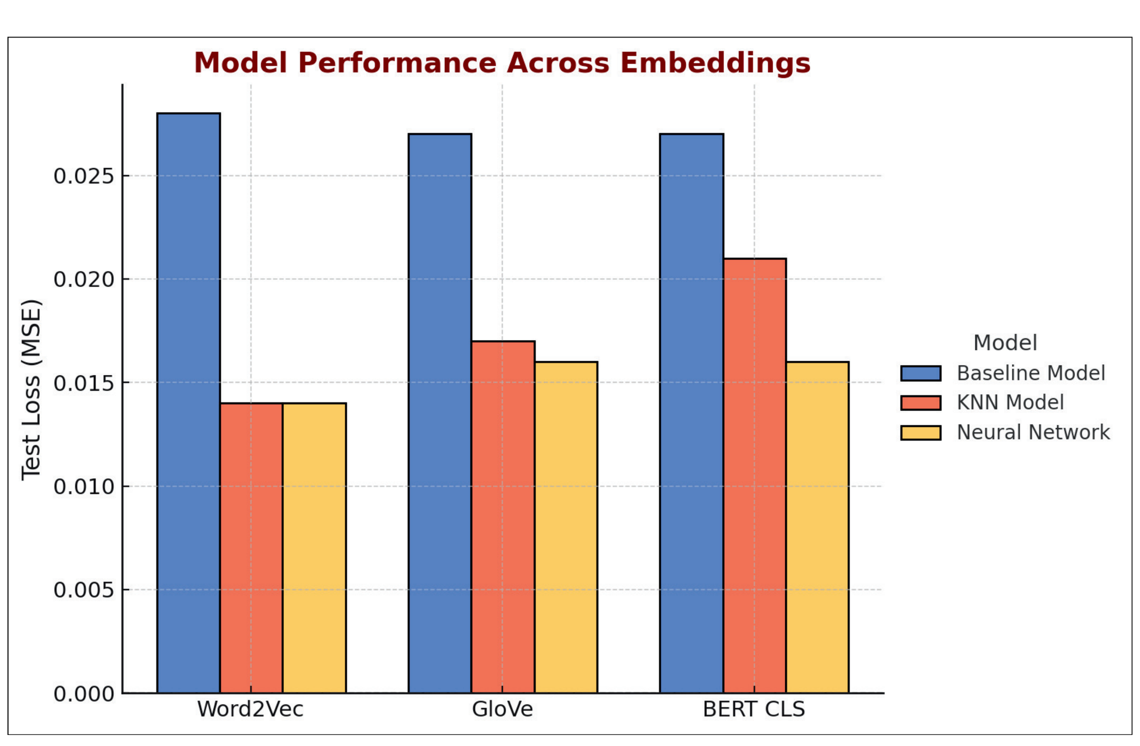
- The dataset contains the strength of words across six perceptual modalities **(touch, hearing, smell, taste, vision, and interoception)** and five action effectors **(mouth, hand, leg, head, and torso)**.
- The data was gathered by surveying **3,500 participants**, who rated the degree to which each word was associated with each modality, using a scale from **0 to 5.**

## 3. Experiment

**Dataset Preparation**

- **Word2Vec-Lancaster:** 33,798 parallel words.
- **GloVe-Lancaster:** 34,346 parallel words.
- **BERT CLS-Lancaster:** 39,855 words (Entire Lancaster Dataset)
- **Intersection Dataset:** Datasets to common words appearing across Word2Vec, GloVe, BERT, and Lancaster for fair comparison.

**Multi-word phrases** were averaged to create a single vector; unmatched phrases were excluded.




**Models**

- **Baseline Model**: Predicts using the mean sensorimotor vector from the training set. Serves as a benchmark.
- **K-Nearest Neighbors (KNN):** Finds 5 most similar examples (cosine similarity) in the training set. Computes weighted averages for predictions.
- **Neural Network:** 1 hidden layer (64/128 neurons).
  - Input sizes: **Word2Vec (300), GloVe (100), BERT CLS (768).**
  - Outputs an 11-dimensional vector for Lancaster Norms.
  - Optimized with **Adam (learning rate: 0.001**) for **10 epochs, batch size: 128**

## 4. Unexpected Sensorimotor Connections: Place Names

Some place names exhibit surprising sensory associations in BERT embeddings. Words like "Napoli" and "Roma" connect to food-related concepts, likely reflecting **cultural and contextual influences** within the dataset.

| Word | Sensorimotor Neighbors |
|---|---|
| napoli | LEMONADE, ORANGE JUICE, EGGNOG, LISTERINE, EDIBILITY |
| roma | CAKE, CREME BRULEE, CROISSANT, MANGO, SWEET PEPPER |
| padua | EGGNOG, MARMALADE, POTPIE, CHEDDAR, MILK CHOCOLATE |
| laval | MARMALADE, ESCARGOT, WASABI, LENTIL, CHARDONNAY |
| lagos | HEADCHEESE, GUMBO, TRUFFLE, SALAD OIL, PARSLEY |

These words were some of the most distant from the **average lancaster embedding,** selected in an attempt to explore the outliers in BERT vocabulary.

The associations reveal interesting patterns. For example:

- Roma aligns with baked goods and sweets, reflecting ties to Italian cuisine.
- Lagos is linked to savory foods, hinting at a culinary context in the data.

These results suggest that place names capture cultural or contextual associations in BERT, especially when trained to learn to **extract sensorimotor information**.

## 5. Emojis and Limitations of Model

**Initial Approach:** Using BERT Base

BERT Base embeddings do not include emojis in the vocabulary.

**Solution**: Replaced emojis with textual descriptions (e.g., 🥭 → "mango").

- Generated meaningful sensorimotor neighbors.
- Results suggest potential for **enhanced emoji prediction**

| Emoji Description | Sensorimotor Neighbor |
|---|---|
| tangerine | PINEAPPLE, APRICOT, RED CABBAGE, BOYSENBERRY, BELL PEPPER |
| face_savoring_food | INDULGE, CONSUMING, NUTRITIOUS, NUTRIENT, NICOTINE |
| call_me_hand | MISCOMMUNICATION, CRITICIZE, INARTICULATELY, ACCUSE, SNARKY |
| drum | CASH REGISTER, XYLOPHONE, FIDDLE, BLOW DRYER, CD PLAYER |

**Retraining with RoBERTa**

- RoBERTa embeddings include emojis in the vocabulary.
- **Results**: Neural Network achieved better test loss **(MSE: 0.012)** than BERT Base.
- Generated **poor sensorimotor neighbors** for emojis:
  - Neighbors consisted of abstract, repetitive words like "a**cceptableness" and "extremeness."**
  - Failed to associate emojis with **concrete sensorimotor properties**.

| Emoji | Sensorimotor Neighbor |
|---|---|
| 🍊 | ACCEPTABLENESS, ORDINARINESS, INCOMPATIBILITY, EXTREMENESS, TANTALIZATION |
| 😋 | CONFORMABLE, ACCEPTABLENESS, DEVELOPMENT, MULTITUDE, US |
| 🤙 | EXPERIMENTALLY, ORDINARINESS, US, MULTITUDE, TANTALIZATION |
| 🥁 | ACCEPTABLENESS, WEDDING ANNIVERSARY, INCOMPATIBILITY, TANTALIZATION, EXTREMENESS |

## 6. Other Downstream Applications and Future Work

- **Potential Applications:**
  - **Sarcasm detection**: Leveraging sensory and physical information to detect nuanced emotional tones in text.
  - **Recommender systems:** Using sensorimotor embeddings to provide personalized content recommendations (e.g., food, music, or activities).
- **Future Work:**
  - Refine embeddings to better **handle multimodal content** (e.g., emojis).
  - Explore **enhanced integration of sensorimotor properties** into downstream NLP tasks.

**Open to suggestions for other possible downstream applications!**